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# Cost-Aware Design of Seawater-Resistant Copper Alloy Powders for Blown-Powder Directed Energy Deposition

**Research problem and objectives.** Navy seawater systems depend on 70-30 copper-nickel and nickel-aluminum bronze (NAB), alloys developed for casting and wrought processing. Directed energy deposition (DED) with blown powder can build large parts and repair damaged ones without a foundry, but its fast, repeated heating and cooling give microstructures these alloys were not designed for. Printed NAB can retain a phase that corrodes selectively in seawater unless heat treated, which is impractical for many repairs made in place. Designing a copper alloy for DED runs through steps whose costs differ by orders of magnitude. A thermodynamic calculation takes minutes, a powder lot a day, a DED session days, and a seawater immersion test weeks. Most campaigns send each candidate through every step, or screen by simulation and lose good candidates when it misses an effect. This project asks how to choose the next measurement and candidate when steps differ in cost and results arrive at different times. We answer it for a pre-alloyed copper alloy powder for single-stream blown-powder DED that deposits without cracks or pores, on itself and on the alloys it would repair, and resists seawater as deposited at least as well as 70-30 copper-nickel. The objectives are to extend a cost-aware search method that learns from partial results to several competing objectives and to many experiments in progress at once, and to give it thermodynamic calculations along the deposit's thermal history. We will demonstrate it on a physical workflow from powder to repair coupon and measure how well a one-hour electrochemical screen predicts immersion tests that take weeks.

**Technical approach.** Calculations decide where experiments go, and experiments correct the calculations. Each step (thermodynamic and heat-flow calculation, powder made in-house by ultrasonic atomization and sized for blown powder, pre-placed powder tracks, DED deposits and repair coupons, microscopy, tensile tests, and seawater corrosion tests) gets its own statistical model. The models are linked in the order of the steps, so uncertainty carries through and a candidate can stop at any step. A cost-aware rule chooses the experiment with the largest expected gain per dollar. A random share of the candidates the screens would stop is tested anyway, to measure how often the screens are wrong. Five hypotheses with stated rejection criteria test the cost savings, the calibration of predictions, process-aware physics, working around pending results, and the corrosion screen. We fix targets before each test and release software, data, and models each year.

**Anticipated outcomes.** If successful, the project will deliver the first cost-aware method of this kind for competing properties and staggered results, and measure how much it saves over full-chain testing and what process-aware physics adds to alloy design. It will also produce corrosion data for DED-deposited copper alloys, with a measured link between a fast screen and long immersion tests, and new copper alloys designed for blown-powder DED, each with its powder specification, deposition settings, and tensile, corrosion, and repair data.

**Impact on Department of War capabilities.** Copper alloys carry seawater through Navy ships and form their propellers. An alloy that can be deposited into new parts or onto worn ones, and that resists seawater without heat treatment, would let shipyards and depots make and repair parts that now wait on castings. The method lowers the cost of qualifying each new composition by running long tests only where they change a decision, and applies to other alloys and processes at Navy laboratories and depots. It will train two doctoral students and undergraduates in powder making, deposition, corrosion testing, and autonomous experimentation.